

# EFC-C: A Thermodynamic Framework for Cognitive Entropy and Psychiatric Biomarkers

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

Research Note v0.1 — March 31, 2026

DOI: [10.6084/m9.figshare.31940505](https://doi.org/10.6084/m9.figshare.31940505)

**Abstract.** We present Energy-Flow Cosmology – Cognition (EFC-C), a thermodynamic extension of the Energy-Flow Cosmology (EFC) framework applied to neural systems. EFC-C models the brain as an open thermodynamic system governed by entropy-gradient dynamics, producing a scalar observable — the total neural entropy production rate  $\dot{S}_{\text{neural}}$  — that serves as a quantitative biomarker of cognitive state. The framework distinguishes itself from the Entropic Brain Hypothesis (EBH) of Carhart-Harris et al. (2014) by predicting not only the magnitude of entropy changes across conscious states, but also their topological structure: the spatial distribution of entropy gradients across large-scale cortical networks. We derive three falsifiable predictions and compare them with published EEG/fMRI data on schizophrenia, major depressive disorder (MDD), and healthy resting-state recordings. Consistency is found across all three comparisons, motivating a prospective clinical validation protocol using open neuroimaging datasets.

## 1 Introduction

The relationship between thermodynamic entropy and conscious experience has a long theoretical lineage, from Shannon’s information entropy to modern neuroscientific applications.

The Entropic Brain Hypothesis (EBH) [1] proposes that the richness of conscious states is indexed by the entropy of spontaneous brain activity, with psychedelic states exhibiting elevated entropy relative to ordinary wakefulness, and disorders such as depression exhibiting pathologically low entropy.

While EBH has attracted substantial empirical support [2, 3], it operates at the level of scalar entropy magnitudes. It does not provide a mechanistic account of why entropy should be spatially heterogeneous across networks, nor does it make topological predictions about the direction and gradient of entropy flow between cortical regions.

Energy-Flow Cosmology (EFC) is a framework originally developed to explain galactic rotation curves and large-scale structure through entropy-gradient dynamics [11]. Its core postulate — that physical complexity emerges from directed entropy gradients, not from entropy magnitude alone — is domain-general. Here we derive EFC-Cognition (EFC-C), the neural instantiation of this principle.

EFC-C makes three testable predictions beyond EBH:

1. **Topological gradient prediction:** Entropy production is highest in integrative hub regions (default mode network, DMN; salience network) and flows directionally toward

primary sensory cortices.

2. **Disorder-specific gradient inversion:** Psychiatric disorders are characterised by gradient inversions — not merely reduced entropy magnitude — with specific signatures distinguishing schizophrenia (excess anterior entropy), MDD (excess posterior entropy), and ADHD (fragmented gradient topology).
3. **Quantitative threshold:** The onset of productive cognitive function corresponds to a minimum entropy production rate  $\dot{S}_{\text{neural}} > \dot{S}^*$ , where  $\dot{S}^*$  is a system-dependent threshold derivable from network connectivity matrices.

## 2 Theoretical Framework

### 2.1 EFC-C Core Postulate

In EFC, the entropy production rate density  $\dot{s}(\mathbf{x}, t)$  drives emergent structure. For a neural system partitioned into regions  $R_i$ , we define the total neural entropy production rate:

$$\dot{S}_{\text{neural}} = \sum_i \int_{R_i} \dot{s}_i dV \quad (1)$$

where  $\dot{s}_i$  depends on metabolic activity, synaptic coupling, and information throughput in region  $i$ . The entropy gradient tensor  $\nabla \dot{S}$  encodes the directional structure of entropy flow across the connectome.

### 2.2 Relation to Established Frameworks

EFC-C situates itself relative to three existing frameworks:

Table 1: Comparison of EFC-C with existing thermodynamic brain theories.

| Framework                 | Magnitude | Topology | Gradient dir. | Psych.   |
|---------------------------|-----------|----------|---------------|----------|
| EBH [1]                   | Yes       | No       | No            | Partial  |
| Free Energy Principle [4] | Implicit  | No       | No            | Indirect |
| IIT [5]                   | No        | Yes      | No            | Indirect |
| EFC-C (this work)         | Yes       | Yes      | Yes           | Direct   |

### 2.3 Entropy Gradient in Psychiatric States

For a healthy resting brain, EFC-C predicts a centrifugal entropy gradient: highest  $\dot{s}$  at DMN hubs, decreasing toward sensory-motor periphery. This gradient sustains the flexible, integrative processing characteristic of normal cognition.

Under the EFC-C model, major psychiatric disorders correspond to characteristic deformations of this gradient:

$$\Delta \nabla \dot{S}_{\text{disorder}} = \nabla \dot{S}_{\text{patient}} - \nabla \dot{S}_{\text{healthy}} \quad (2)$$

Eq. (2) is in principle measurable from resting-state fMRI or high-density EEG using multiscale sample entropy (MSE) decomposition.

### 3 Predictions and Empirical Consistency

#### 3.1 Prediction P1: Topological Entropy Structure

EFC-C predicts that the spatial distribution of brain entropy across resting-state networks follows a hub-to-periphery gradient. This is consistent with published fMRI findings showing that entropy-related measures (Lempel–Ziv complexity, MSE at coarse scales) are highest in association cortices and DMN regions [3, 10].

#### 3.2 Prediction P2: Disorder-Specific Gradient Signatures

Published EEG literature using multiscale fuzzy entropy (MFE) provides direct evidence for disorder-specific entropy topologies:

Table 2: EFC-C predicted entropy gradient signatures vs. observed EEG/fMRI findings.

| Condition     | EFC-C prediction                                   | Observed (literature)                                                             | Ok? |
|---------------|----------------------------------------------------|-----------------------------------------------------------------------------------|-----|
| Schizophrenia | Elevated anterior entropy, fragmented hub gradient | Increased frontal MFE, prefrontal–temporal dysconnection [7]                      | ✓   |
| MDD           | Reduced entropy magnitude, posterior bias          | Lower MSE in left frontal, reduced EEG microstate complexity [8]                  | ✓   |
| Healthy       | Centrifugal gradient, maximal DMN entropy          | Brain entropy positively correlated with intelligence in association cortices [9] | ✓   |

#### 3.3 Prediction P3: Entropy Threshold for Cognitive Function

EFC-C predicts a threshold  $\dot{S}^*$  below which coordinated cognition degrades. This is consistent with disorders-of-consciousness research, where loss of Lempel–Ziv complexity below a critical level reliably predicts unresponsive states [2].

#### 3.4 Key Distinction from EBH

The Entropic Brain Hypothesis correctly identifies entropy magnitude as an index of conscious state richness but does not predict the spatial topology of this entropy. A critical commentary on EBH noted that its configurational entropy “establishes a connection between the topography of the energy landscape and the system’s dynamics” only if the coarse-graining level is carefully chosen [6].

EFC-C provides the missing mechanistic layer: entropy topology is not an epiphenomenon but a consequence of directed energy-gradient dynamics. This implies that EBH and EFC-C are not competing — EBH is the scalar projection of the more complete EFC-C tensor field.

## 4 Discussion

### 4.1 Theoretical Scope

EFC-C formalises the intuition that the brain, like cosmological large-scale structure in EFC, organises itself through entropy-gradient flows. The same mathematical structure — a scalar entropy production field with a directional gradient — that predicts galaxy rotation anomalies without dark matter [11] also predicts the topological signatures of psychiatric states.

This cross-domain consistency is not coincidental under the EFC framework: it reflects the domain-generalty of entropy-gradient dynamics as an organising principle for complex open systems.

### 4.2 Limitations

Several caveats apply to the present theoretical note:

1. **No direct model fit:** The comparisons in Table 2 are qualitative consistency checks, not likelihood-ratio tests. EFC-C has not yet been fit to a neural dataset.
2. **Parameter underdetermination:** The mapping from EFC cosmological parameters to neural entropy production rates requires a bridge theory that has not yet been derived formally.
3. **Gradient measurement:** Empirically extracting  $\nabla \dot{S}_{\text{neural}}$  from EEG or fMRI requires careful choice of entropy estimator, scale, and parcellation scheme.

### 4.3 Proposed Next Steps

1. Fit the EFC-C gradient model to open resting-state fMRI data (e.g. Human Connectome Project, OpenNeuro schizophrenia datasets) using MSE parcellation.
2. Derive the formal bridge from EFC cosmological entropy-gradient equations (Eq. 1 in [11]) to neural  $\dot{s}_i$ .
3. Test P2 differential predictions: schizophrenia vs. MDD classification using the predicted gradient signatures as features.
4. Compare AIC against pure-EBH (scalar entropy) model on the same data.

## 5 Reproducibility

This theoretical note does not include original data analysis. All empirical claims are referenced to published peer-reviewed literature.

This document is archived at:

[doi:10.6084/m9.figshare.31940505](https://doi.org/10.6084/m9.figshare.31940505)

The EFC cosmological framework underlying EFC-C is documented at:

[doi:10.6084/m9.figshare.31224538](https://doi.org/10.6084/m9.figshare.31224538)

The EFC codebase and research materials are available at:

[github.com/supertedai/EFC](https://github.com/supertedai/EFC)

## References

- [1] Carhart-Harris, R.L., Leech, R., Hellyer, P.J., et al. (2014). The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs. *Frontiers in Human Neuroscience*, 8, 20. [doi:10.3389/fnhum.2014.00020](https://doi.org/10.3389/fnhum.2014.00020)
- [2] Carhart-Harris, R.L. (2018). The entropic brain — revisited. *Neuropharmacology*, 142, 167–178. [doi:10.1016/j.neuropharm.2018.03.010](https://doi.org/10.1016/j.neuropharm.2018.03.010)
- [3] Rosas, F.E., et al. (2023). A whole-brain model of the neural entropy increase elicited by psychedelic drugs. *Scientific Reports*, 13, 6615. [doi:10.1038/s41598-023-32649-7](https://doi.org/10.1038/s41598-023-32649-7)
- [4] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138.
- [5] Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.
- [6] Papo, D. (2016). Commentary: The entropic brain. *Frontiers in Human Neuroscience*, 10, 423. [doi:10.3389/fnhum.2016.00423](https://doi.org/10.3389/fnhum.2016.00423)
- [7] Noroozi, A., et al. (2025). Machine learning-based differentiation of schizophrenia and bipolar disorder using multiscale fuzzy entropy and relative power from resting-state EEG. *Translational Psychiatry*, 15, 120. [doi:10.1038/s41398-025-03354-y](https://doi.org/10.1038/s41398-025-03354-y)
- [8] Pan, S., et al. (2025). Novel EEG-based diagnostic framework for major depressive disorder using microstate and entropy features. *Cognitive Neurodynamics*. [doi:10.1007/s11571-025-10305-2](https://doi.org/10.1007/s11571-025-10305-2)
- [9] Oxford Neuroscience of Consciousness (2023). Psychedelics, entropic brain theory, and the taxonomy of conscious states. *Neuroscience of Consciousness*, 2023(1), niad001. [doi:10.1093/nc/niad001](https://doi.org/10.1093/nc/niad001)
- [10] Creel, W.T. and Hartman, R.E. (2025). EEG entropy modulation as a biomarker of emotion regulation and resilience. *IBRO Neuroscience Reports*.
- [11] Magnusson, M. (2026). Energy-Flow Cosmology: SPARC Validation Phase 3. Figshare Research Note. [doi:10.6084/m9.figshare.31224538](https://doi.org/10.6084/m9.figshare.31224538)